Chapter 1

Analytical and numerical modelling of wave propagation in (meta)poro(elastic) layers

This chapter lays the foundation for the thesis by providing the fundamental theoretical notions and methods used in this work, in order to guide the reader through the various chapters that will follow. At first, an overview of the acoustic modelling of porous media is proposed. Following the presentation of the state of art in the introduction and of the areas of focus that have been identified from the latter, the key notions and methods for the computation of the scattering of waves by metaporous structures will be presented. From there, the formalism for the calculation of dispersion curves will be presented along with an overview of the methods used for such computations. Both of these topics will be illustrated with preliminary results in simple configurations.

1.1 Modelling of porous media

Porous media is a category of materials which encompasses naturally occuring substances such as rocks, soils and biological tissues, and engineered materials like ceramics, metallic foams, rockwool or certain types of polymers. They are considered biphasic, that is constituted of two distinct domains: a pore network $\mathbf{\Omega}_f$ filled by a saturating fluid embedded in an elastic matrix $\mathbf{\Omega}_s$. A diagram of this microscopic structure is shown in Fig. 1.1. The porosity is defined as the fraction of fluid over the total volume of the material, as $\phi = V(\Omega_f)/V(\Omega)$. In the limit case, $\phi = 0$ the material becomes purely a solid while at $\phi = 1$, it becomes a fluid. At microscopic scales, the geometry of the fluid phase thus forms a complex microstructure. Fortunately, at airborne acoustic scales, these materials can often be considered at a meso-/macroscopic level and therefore models have been developed to describe the large-scale motion of these materials. In this context, they can be regarded as homogeneous and described by a set of effective parameters that accounts for their complex geometry. The porous structure of these materials causes the incoming acoustic energy to be dissipated through friction (viscous losses) along the duct/pore network, while thermal exchanges and elastic dissipation take place when the skeleton and the fluid phase are coupled. These losses are thus dependant on the geometry, which parametrized by the porosity, the saturating fluid properties, the pore network size, connectivity and shape, as well as the frame mechanical parameters.

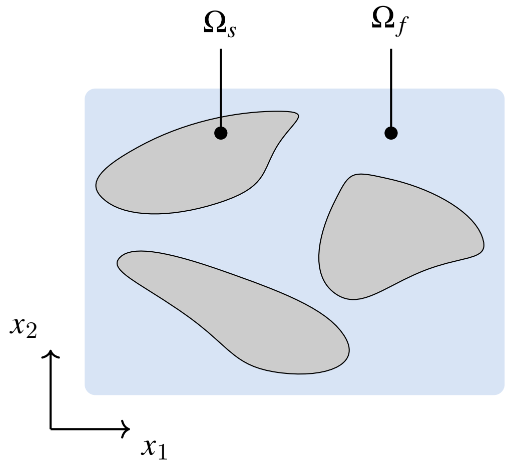

Fig. 1.1: Porous material structure

1.1.1 Rigid-frame approximation

In the so-called rigid-frame approximation, the skeleton of the porous material is considered motionless. Therefore, the pore walls are rigid walls and the material can be modeled as an *equivalent fluid* which possesses complex effective properties in order to model the losses mechanisms. A harmonic acoustic propagation is assumed with a positive time convention $e^{\mathrm{i}\omega t}$ troughout this manuscript (with the exception of Chapter **??**). The Helmholtz equation governs the propagation,

$$(\nabla^2 + k_{eq}^2)p(\boldsymbol{x}) = 0, \tag{1.1}$$

where p is the pressure field and k_{eq} is the wavenumber of the equivalent fluid. In the case where porous or fibrous materials have a porosity close to 1 and the frequency range of interest is limited, the Delany-Bazley model was originally developed in 1970 to propose empirical expressions for the values of the complex wavenumber and characteristic impedances[1,2] and later improved by Miki[3]. Although limited in accuracy and validity domain, these models has been widely used since it only required the flow resistivity of the porous material. Later on, a more precise semi-phenomenological model additionally accounting for the viscous and inertial dissipative effects was developed from 1987 to 1991. It is the Johnson-Champoux-Allard (JCA) model which models the complex effective density $\rho_{eq}(\omega)$ accounts for the frequency dependant viscous losses and effective bulk modulus of the material K_{eq} accounting for thermal exchanges in the layer as,

$$\begin{aligned}
\rho_{\mathrm{eq}}(\omega) &= \frac{\rho_f \alpha_\infty}{\phi}\left(1 + \frac{\mathrm{i}R_f\phi}{\rho_f\alpha_\infty\omega}\sqrt{1 - \mathrm{i}\omega\rho_f\mu_f\left(\frac{2\alpha_\infty}{R_f\phi\Lambda}\right)^2}\right), \\
K_{\mathrm{eq}} &= \frac{\kappa P_0}{\phi}\left(\kappa - (\kappa-1)\middle/\left(1 + \frac{\mathrm{i}R_f\phi}{\rho_f\alpha_\infty \mathrm{Pr}\omega}\sqrt{1 - \mathrm{i}\mathrm{Pr}\omega\rho_f\mu_f\left(\frac{2\alpha_\infty}{R_f\phi\Lambda'}\right)^2}\right)\right)^{-1}.
\end{aligned} \tag{1.2}$$

These expressions involve a large number of effective parameters for the porous material; α_∞ is the tortuosity, R_f is the flow resistivity, Λ and Λ' are respectively the viscous and thermal characteristic length. The saturating fluid is characterized in this model by its heat capacity ratio κ (commonly also denoted γ), density ρ_f, kinematic viscosity μ_f, pressure P_0 and Prandtl number $\mathrm{Pr} = c_p\mu_f/k$ with c_p is the specific heat and k the thermal conductivity. We can introduce also the Biot frequency $\omega_c = R_f\phi/(\rho_f\alpha_\infty)$ that defines the transition frequency between the inertial and viscous regimes.

1.1.2 Elastic motion of the skeleton: Biot theory

While the equivalent fluid formalism is able to accurately model porous material in specific sets of conditions, that is when the skeleton can be assumed to be motionless, this assumption can quickly become limited as soon as the said skeleton is subjected to an elastic deformation. This is can be the case when the saturating fluid of the PEM has a high density (e.g. water), when the skeleton is coupled to elastic plates or when the acoustic radiation has a wavelength comparable to the pore scale, although these are just examples. In this situation, the full elastic motion should be modeled, so that to account for the coupling between the solid matrix and its fluid filling. The Biot theory[4–7]is a formalism that encompasses the theory of the deformation of porous material saturated by an incompressible fluid, used both in geophysics and acoustics. For sake of conciseness, the various models will be briefly be presented along with the relevant physical quantities. A full derivation would be out of scope and is presented in existing works[2].

Before presenting the full Biot model, some quantities should be provided additionally to the equivalent fluid parameters. The frame is assumed to undergo elastic deformations, thus the common mechanical parameters used are the solid density of the frame material ρ_s as well as its apparent density $\rho_1 = (1-\phi)\rho_s$. The bulk modulii are denoted K_b and K_s, respectively for the frame and the material. The shear modulus is also considered complex, accounting for structural losses as $\widetilde{N} = N(1+\mathrm{i}\eta)$ with η the loss factor. Symbols written with a tilde denote complex expressions and therefore account for some sort of loss mechanisms. For the fluid, the apparent density is thus $\rho_2 = \phi\rho_f$ and the inertial interaction density defined by $\rho_{12} = -\phi\rho_f(\alpha_\infty - 1)$. A summary on Biot theory is also proposed in Chap. **??**) with the tilde symbols omitted for ease of reading.

Formulations for the motion equations

The Biot poroelastic equations from the original 1956 formulation[4,6,7] depend on the solid displacement vector $\boldsymbol{u}_s$ and the fluid displacement $\boldsymbol{u}_f$,

$$\begin{aligned} \boldsymbol{\nabla}\cdot\boldsymbol{\sigma}_s(\boldsymbol{u}_s,\boldsymbol{u}_f) &= -\omega^2\widetilde{\rho_{11}}\boldsymbol{u}_s - \omega^2\widetilde{\rho_{12}}\boldsymbol{u}_f, \\ \boldsymbol{\nabla}\cdot\boldsymbol{\sigma}_f(\boldsymbol{u}_s,\boldsymbol{u}_f) &= -\omega^2\widetilde{\rho_{12}}\boldsymbol{u}_s - \omega^2\widetilde{\rho_{22}}\boldsymbol{u}_f, \end{aligned} \tag{1.3}$$

where $\boldsymbol{\sigma}^s$ is the solid stress tensor, $\boldsymbol{\sigma}^f$ the fluid stress tensor and the apparent densities of the solid-phase is $\widetilde{\rho_{11}}$, that of the fluid is $\widetilde{\rho_{22}}$ and the coupling apparent density is $\widetilde{\rho_{12}}$. They are complex quantities expression that also account for the contribution of viscous forces and are given as,

$$\widetilde{\rho_{22}} = \phi^2\widetilde{\rho_{eq}}, \quad \widetilde{\rho_{12}} = \rho_2 - \widetilde{\rho_{22}}, \quad \widetilde{\rho_{11}} = \rho_1 - \widetilde{\rho_{12}}. \tag{1.4}$$

These coupled motion equations are associated to the stress-strain relation,

$$\begin{aligned} \boldsymbol{\sigma}_{s,ij} &= \left[(\widetilde{P} - 2\widetilde{N})\nabla\cdot\boldsymbol{u}_s + \widetilde{Q}\nabla\cdot\boldsymbol{u}_f\right]\delta_{ij} + 2\widetilde{N}\boldsymbol{\varepsilon}_{s,ij}, \\ \boldsymbol{\sigma}_{f,ij} &= \left(\widetilde{Q}\nabla\cdot\boldsymbol{u}^s + \widetilde{R}\nabla\cdot\boldsymbol{u}_f\right)\delta_{ij} = -\phi p, \end{aligned} \tag{1.5}$$

where $\nabla\cdot\boldsymbol{u}_s$ and $\nabla\cdot\boldsymbol{u}^f$ correspond to the dilatation of the solid and fluid phases respectively and $\boldsymbol{\varepsilon}_s$, is the solid-phase strain tensor. The Biot coefficients are denoted as $\widetilde{P}$ for the solid elastic coefficient, $\widetilde{Q}$ for the elastic-fluid coupling coefficient, and $\widetilde{R}$ for the fluid-phase coefficient. In the context of a PEM saturated by a light fluid, their expressions are

$$\widetilde{P} = K_b + \frac{4}{3}N + (1-\phi)^2\frac{\widetilde{K_{eq}}}{\phi}, \quad \widetilde{Q} = \widetilde{K_{eq}}(1-\phi), \quad \widetilde{R} = \phi\widetilde{K_{eq}}. \tag{1.6}$$

The full coefficients are generalized forms required when the limit $K_b/K_s \to 0$ no longer holds, as is the case for water-saturated PEM.

A different formulation, especially convenient for finite-elements discretization is known as the $(\boldsymbol{u}^s, p)$ formulation. First, new parameters in terms of the Biot elastic coefficients and apparent densities can be defined,

$$\widehat{A} = \widetilde{P} - 2\widetilde{N}_s - \frac{\widetilde{Q}^2}{\widetilde{R}}, \quad \widehat{\rho} = \widetilde{\rho_{11}} - \frac{\widetilde{\rho_{12}}^2}{\widetilde{\rho_{22}}}, \quad \beta = \phi\left(\frac{\widetilde{\rho_{12}}}{\widetilde{\rho_{22}}} - \frac{\widetilde{Q}}{\widetilde{R}}\right). \tag{1.7}$$

β is another coupling coefficient, also denoted as $\widetilde{\gamma}$ in the literature. Using the second row of Eq. (1.3) allows for the expression of the fluid displacement as, $\boldsymbol{u}^f = \phi/(\widetilde{\rho_{22}}\omega^2)\nabla p - \widetilde{\rho_{12}}/\widetilde{\rho_{22}}\boldsymbol{u}^s$ and using that in Eq. (1.3) and Eq. (1.5) allow to rewrite the coupled motion equations. Using these, the so-called $(\boldsymbol{u}_s, p)$ formulation of the Biot theory[8] is expressed as,

$$\begin{aligned} \nabla \cdot \widehat{\boldsymbol{\sigma}}_s + \widehat{\rho}\omega^2 \boldsymbol{u}_s + \beta \boldsymbol{\nabla} p = 0, \\ \frac{\nabla^2 p}{\rho_{22}\omega^2} - \frac{\beta}{\phi^2}\nabla \cdot \boldsymbol{u}_s + \frac{1}{R}p = 0, \end{aligned} \tag{1.8}$$

where $\widehat{\boldsymbol{\sigma}}_s$ is the *in vacuo* solid stress tensor,

$$\widehat{\boldsymbol{\sigma}}_{s,ij} = \boldsymbol{\sigma}_{s,ij} - \phi\frac{\widetilde{Q}}{\widetilde{R}}p\delta_{ij} = \widehat{A}\nabla \cdot \boldsymbol{u}_s\delta_{ij} + 2N_s\boldsymbol{\varepsilon}_s \tag{1.9}$$

Propagation in unbounded poroelastic media

These various formulations provide several way of modelling equivalently wave propagation in poroelastic media. It appears that PEM supports three waves: a compression φ_p and shear wave ψ_s related to the solid phase as well as an additional compression wave φ_a, related to the fluid phase. In the case of each of these formulations, the fields can be expressed depending on these potentials,

$$\boldsymbol{u}_s = \nabla\varphi_p + \nabla\varphi_a + \nabla \times \psi_s, \qquad \boldsymbol{u}_f = \mu_p\nabla\varphi_p + \mu_a\nabla\varphi_a + \mu_s\nabla \times \psi_s, \tag{1.10}$$

with μ_p, μ_a, μ_s the solid displacement over the fluid displacement ratio coefficients. The wavenumbers for each of these waves are, following section 6.5 of Allard and Atalla textbook[2] and denoted as $\delta_i, i = 1, 2, 3$ in the text,

$$\begin{aligned} k_{p,a} = \sqrt{\frac{1}{2}\left(\delta_2^2 + \delta_{eq}^2 \pm \delta_t\right)}, \\ k_s = \omega\sqrt{\frac{\widehat{\rho}}{\widetilde{N}}}, \end{aligned} \tag{1.11}$$

respectively for the solid compression, fluid compression and shear wave, with,

$$\delta_{eq} = \omega\sqrt{\frac{\widetilde{\rho}_{eq}}{\widetilde{K}_{eq}}}, \quad \delta_1 = \omega\sqrt{\frac{\widehat{\rho}}{\widehat{P}}}, \quad \delta_2 = \omega\sqrt{\frac{\widehat{\rho_s}}{\widehat{P}}}, \quad \delta_t = \sqrt{(\delta_2^2 + \delta_{eq}^2)^2 - 4\delta_{eq}^2\delta_1^2}. \tag{1.12}$$

The propagation characteristics, i.e. wavenumbers of the waves propagating in porous and poroelastic media are illustrated in Figure 1.2. As discussed in the previous sections, the wavenumber for the equivalent fluid as well as the three for the poroelastic layer are complex. Their real part account for the propagation of the phase flow, inversly proportional to the wave velocity and their imaginary part accounts for the attenuation that occurs due to the losses inherent to each of these materials. It appears that at low frequencies, the attenuation is very low for all waves. The velocity of the compression and shear waves (observed on the figures by the slope of the real parts of the wavenumbers) is much higher than the waves in the fluid for the equivalent fluid and the compression fluid phase velocity, as they propagate in a soft solid where the velocity is usually much higher than in a fluid.

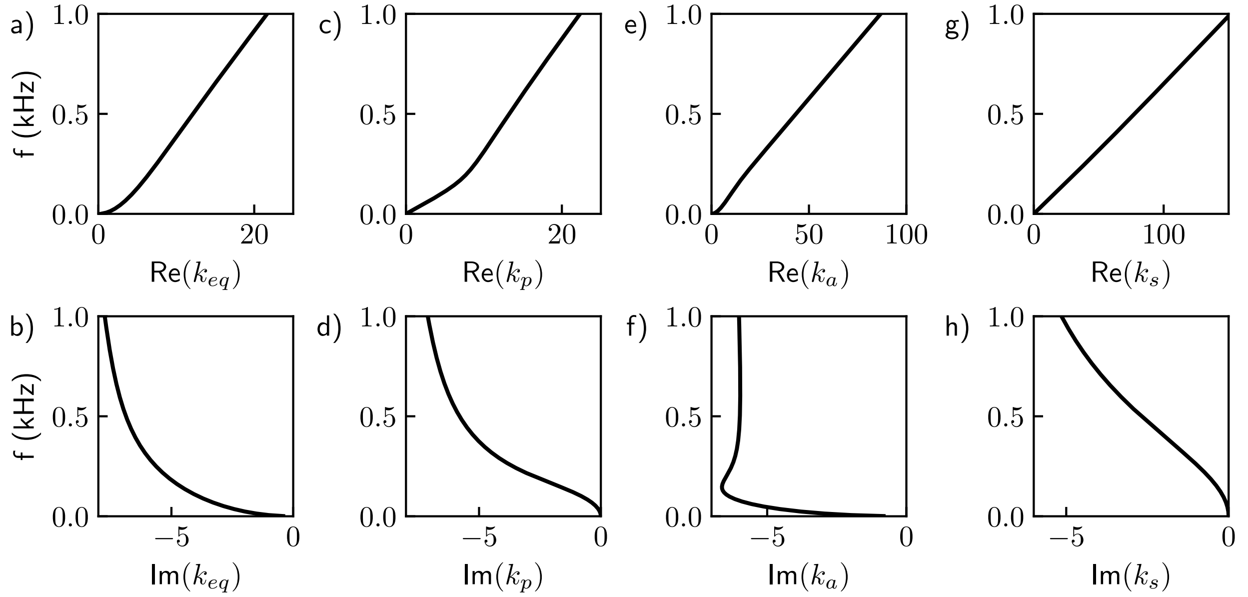

Fig. 1.2: Wavenumbers of waves in unbounded porous media in the case of an a-b) equivalent fluid and c-h) for the Biot model.

1.2 Modelling of the wave propagation in homogeneous layers

In the previous section, a formulation for rigid-frame porous media and two formulations for poroelastic media modelling have been presented. Concluding the presentation of these motion equations, some characteristics of the wave propagation in unbounded porous and poroelastic media have been shown. The present section proposes the introduction of a geometry, that is to say boundaries or interface conditions delimiting different domains characterized each by a different type of material. Generally, the configurations studied in this work assume a monochromatic plane wave originated from a fluid half-space, i.e. air, which impinges at the surface of a structure and is scattered (reflected and transmitted) through each interface. If the material in which the wave propagates is bounded by two parallel, free, and flat surfaces, i.e., a plate, then it is refered to as a layer. The waves supported by the medium propagate by reflecting alternatively from one surface to the other; they are said to be guided along the layer. These observations of successive reflections and refractions lead to the concept of waveguides, but their analytical descriptions often leads to writing complex system that can be difficult to solve. To determine the modes of a structure, one must seek to solve the propagation equations and the boundary conditions imposed at its boundaries. In a first section, the scattering of a plane wave by a homogeneous porous layer is investigated along with the various method available. Next, the methods for the computation of the modes of these configurations are presented. In a subsequent section, more complex geometries, those of metaporous structures, are of interest and both methods for the computation of the scattering coefficients and the modes of those configurations will be discussed.

1.2.1 Global Method: plane wave modelling

The analytical description of the system amounts finally to a linear system built from the interface conditions, which unknown vector are the amplitudes of each of the existing waves in the system. The forcing right-hand side corresponds to the incident plane. The solution of the system yields all unknown amplitudes of the system, in particular the reflection and transmission coefficients.

Introducing a simple problem, that of the scattering of an incident plate by a porous layer, we present an example of plane wave modelling for a simple system. The system is 2D with coordinates $\boldsymbol{x} = (x_1, x_2)$ constituted of a porous layer with thickness h and thus two interfaces of infinite extent along x_1, there is a fluid-porous interface at $x_1 = h$ and a porous-fluid interface at $x_1 = 0$. Both media of the air half-spaces and porous layer are modeled as fluid, thus the interfaces conditions should impose continuity of the pressure fields and normal particle displacements.

Assuming a $e^{\mathrm{i}\omega t}$ time-dependance, the incident pressure field originated from the upper domain takes the form of,

$$p_{0+} = a_+ e^{-ik_1x_1} e^{-ik_{2,0}(x_2-h)} + a_- e^{-ik_1x_1} e^{ik_{2,0}(x_2-h)}. \tag{1.13}$$

The pressure field in the porous layer is expressed as a sum of forward and backward waves,

$$p_p = b_+ e^{-ik_1x_1} e^{-ik_{2,eq}x_2} + b_- e^{-ik_1x_1} e^{ik_2(x_2-h)}, \tag{1.14}$$

and that in the lower air domain is,

$$p_{0-} = c e^{-ik_1x_1} e^{ik_{2,0}x_2}, \tag{1.15}$$

with wavenumbers such that they hold for the Snell-Descartes law that states $k_1^2 + k_{2,0}^2 = k_0^2$ and $k_1^2 + k_{2,eq}^2 = k_{eq}^2$ The displacement fields are additionally expressed using the momentum equation relating the latter with the pressure fields as $\partial_2 p = -\mathrm{i}\omega\rho_i u_2, ::i =.$

$$p^t(h) = p^l(h), \quad u_2^t(h) = u_2^l(h) \quad p^l(0) = p^b(0) \quad {}^l_2(0) = u_2^b(0) \tag{1.16}$$

Imposing the interface conditions leads to the following matrix system,

$$\begin{pmatrix} 1 & e^{\mathrm{i}k_{2,eq}h} & 0 & -1 \\ \dfrac{k_{2,eq}}{\rho_{eq}} & -\dfrac{k_{2,eq}}{\rho_{eq}} e^{\mathrm{i}k_{2,eq}h} & 0 & \frac{k_{2,0}}{\rho_0} \\ e^{\mathrm{i}k_{2,eq}h} & 1 & -1 & 0 \\ \frac{k_{2,eq}}{\rho_{eq}} e^{\mathrm{i}k_{2,eq}h} & -\frac{k_{2,eq}}{\rho_{eq}} & -\frac{k_{2,0}}{\rho_0} & 0 \end{pmatrix} \begin{pmatrix} b_+ \\ b_- \\ a_+ \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ a_- \\ \frac{k_{2,eq}}{\rho_{eq}} a_- \end{pmatrix}. \tag{1.17}$$

vérifier l'écriture du système. Imposing the wavenumber $k_1 = k_0 \sin\theta, k_2 = -k_0 \cos\theta$, with θ the angle of incidence, allows to easily solve the system and obtain all scattered amplitudes b_+, b_-, a_+ and c. In the case of a rigidly-backed layer, the interface at $x_2 = h$ is replaced by a rigid termination, that is $u_2(h) = 0$ and thus the terms depending on c vanishes, since p_{0-} does not exist; this reduces the size of the system. In this case, the reflection coefficient R corresponds to the amplitude a_+.

A few results for the simple cases of the scattering of a plane wave by a rigidly-backed layer and in a symmetric configurations are represented by the diagrams in Figs 1.3a and b and results in Figs 1.3c and d, respectively. The GM results are presented by the continuous lines. There, the layer for both porous and PEM are both $h = 5$ cm thick and the incident wave propagates with a $\theta = 30^o$ angle. The parameters for each material corresponds to the one used in Table **??** from Chapter **??**. The system used to obtain the reflection coefficient for the PEM layer is done in a similar fashion as the equivalent fluid and can be read in the appendices of[9].

The first case, also referenced to as a reflection problem compares the equivalent fluid and the full Biot model response. The two curves follow the same trend but additional effects due to the skeleton motion can be observed on the continuous lines while the equivalent fluid response shows to be smoother. For all configurations, the melamine layer is acoustically transparent at lower frequency and absorption can be noticed at slightly higher frequencies.

Generally, the application of the Global Method allows to write the propagation of plane waves through any stack of layer, by putting together a matrix system that contains all interface conditions in the system, in the general form,

$$\boldsymbol{M}(\omega)\boldsymbol{x} = \boldsymbol{f}, \tag{1.18}$$

1.2.2 Transfer Matrix Method

The widely used method we know today as the Transfer Matrix Method (TMM), was initially developed by the works of Thomson**thomson1950** and Haskell**haskell1951** in order to describe elastic wave propagation in multilayer systems. This method can be applied to a range of wave physics among geophysics, quantum mechanics, electromagnetics and obviously acoustics. This method allows for an easy multiphysics coupling with interface matrices leads to large systems.

The core of the method is in the evaluation of the state vector that regroups the variables of interest in the system and evaluate them at each interface. propagate the properties stored in the state vector from one layer to the next. An amount of N layers of the same nature are easy to manipulate since $\boldsymbol{T}_{\text{global}} = \prod_{n=0}^{N} \boldsymbol{T}_n$. This method was deep dived in several textbooks among which we propose Refs.[2,10]. It allows for the computation of complex system assembly of elements by introducing additional elements.[11]. While very versatile, it remains unstable in limit cases at high frequencies or for large thickness of the layers. While the formulation is mathematically exact, the evaluation of the system involves involves finite-precision calculation which leads the very sensitive exponential terms to diverge in the previously specified regimes.

$$\begin{pmatrix} p(0) \\ u_2(0) \end{pmatrix} = \begin{pmatrix} \cos(k_{eq}h) & \mathrm{i}Z_{eq}\sin(k_{eq}h) \\ \mathrm{i}Z_{eq}^{-1}\sin(k_{eq}h) & \cos(k_{eq}h) \end{pmatrix} \begin{pmatrix} p(h) \\ u_2(h) \end{pmatrix} \tag{1.19}$$

From the evaluation of the transfer matrix of the global system, the scattering matrix can be obtained and thus the reflection and transmission coefficients.

The absorption coefficient for the same two configurations as the last part is presented in Figs 1.3c and d with the markers. It can be seen that the same results are obtained.

1.2.3 Guided waves for dispersion problems

From the previously obtained linear systems in Section 1.2.1 for the configurations, and removing the incident wave from the system, i.e. the forcing (right-hand side) term, an homogeneous linear system is obtained, representing the guiding of waves in the layer. This set of equations becomes singular when its determinant vanishes for a given parameter, allowing to identify the modes of the structure. Several formalism can be adopted: solving for frequencies while imposing the wavenumber or the other way around. The first one is useful when interested in the time-dependant effects the structure has on the propagation. In this work, we are interested in the losses generated by the structure and the material and as such frequency is fixed in order to solve wavenumbers.

Computating the modes with semi-analytical tools

for porous/poroelastic materials, root-finding[12] + groby2008 single homogeneous layers

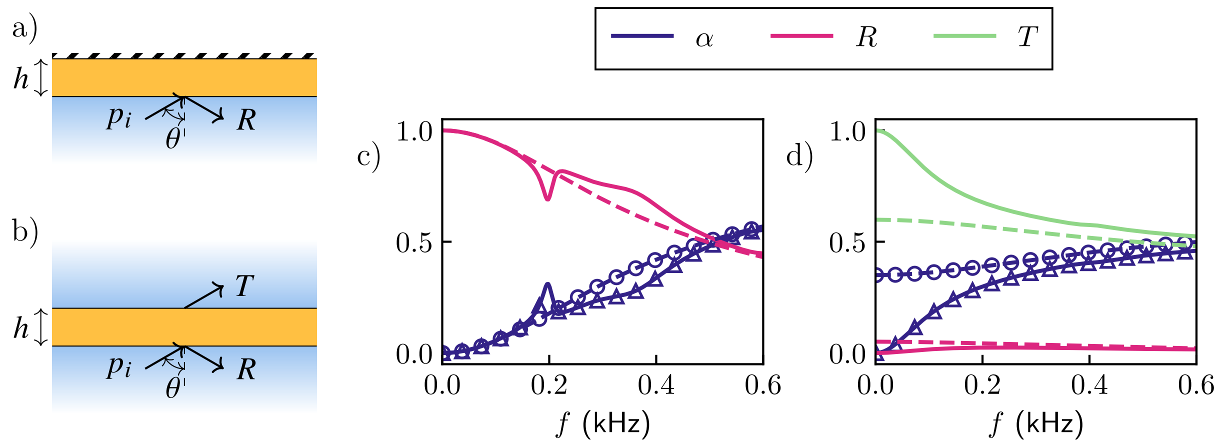

Fig. 1.3: Diagram for the a) reflection, b) transmission problem. Scattering coefficients from a poroelastic layer (continuous) and equivalent fluid layer (dashed) computed using the GM for the c) reflection, d) transmission problem. Results computed with the TMM are also shown for the absorption coefficient for the equivalent fluid (○) and for the absorption coefficient (△)

Once the linear system is obtained, we are faced with either or two of these possibilities. Solving wavenumbers can be done through the writing and solving of a (possibly generalized) eigenvalue problem or by root-finding of solutions in the case where writing the first one appears impossible.

Using Eq. (1.18), root finding such that $\det(\boldsymbol{M}(k_1, \omega)) = 0$; the determinant of the matrix from Eq. 1.17 can be expressed as

$$2\frac{k_{2,eq}k_{2,0}}{\rho_{eq}\rho_0}\cos(k_{2,eq}h) + \mathrm{i}\left(\frac{k_{2,eq}^2}{\rho_{eq}^2} + \frac{k_{2,0}^2}{\rho_0^2}\right)\sin(k_{2,eq}h) = 0 \tag{1.20}$$

Numerics for mode computations

Regarding the computation of complex modes in homogeneous layer, several numerical methods have been investigated in the last decades. Guided waves represents a large topic and we focus here on the case of simple geometries; similar configurations are considered, an unbounded single (or stack of) layers with two infinite interfaces as presented in the section where we were interested in their acoustic response.

Spectral methods have been around for quite some time and have been used for many numerical applications. We focus for this work on Spectral Collocation Method (SCM), which has been especially used for the computation of modes for guided waves in layers.[13]

In the case of simple 2D geometries, a 1D discretization along the transverse direction associated to a harmonic motion in the longitudinal direction. Even if the use of finite elements seems farfetched, this method could be relevant in contexts where boundary effects or guide with varying cross-sections are studied. However, a variation of finite elements based on a 1D discretization of the waveguide is entitled Semi-analytical Finite Elements (SAFE) and is akin to SCM which we have described in the previous paragraph. The 1D discretization is done along the thickness of the layer by regular nodes, and the fields are interpolated using finite elements shape functions. Examples of Semi-analytical Finite Elements (SAFE) use cases are very similar to SCM and include lister des publis ici. Examples are also given in Chapter **??**. A detailed description of the calculation of Lamb modes using a simple SAFE implementation using high-order Lobatto polynomials is proposed in Appendix. **??**. There, its performance is also compared to that of SCM.

discuss surface and guided waves in a paragraph

Fig. 1.4: Discretization for each method

A bestiary of the wave solutions in dissipative and open waveguides

In conservative and closed systems, propagating waves are either guided or trapped waves, while in open non-dissipative systems, the acoustic energy is able to radiate outside of the waveguide. Part of propagating waves are thus attenuated and radiate energy to the open domain. They are thus considered quasi-guided since they are propagating waves with some attenuation. Trapped modes can still occur. New family of modes appear as leaky backward and leaky forward waves that exhibit unique properties. Their counterparts are incoming waves.

The addition of dissipation leads to the mixing of several loss mechanisms: the leakage of waves and the dissipations within the guide (which can be of whatever nature). In this case, the separation between incoming and leaky waves is not possible anymore. This particular aspect has been very scarcely tackled in the literature: an unpublished paper is interested in an elastic waveguide[14] while in optics, a paper discusses the waves guided in nanospheres with a complex permittivity**<empty citation>**vérifier la citation.

There is a discussion to be made on the relevance of the imposition of a local coupling contrary to a full modelling of the surrounding media with some termination condition satisfying the Sommerfeld condition, PML or analogous.

1.2.4 Numerics for the computation of dispersion relations

- presentation of scm, safe + possible with fem
- cite chap. 2

1.3 Methods for modelling metaporous structures

1.3.1 Scattering of waves by periodic structures

As has been studied in the introduction, a large literature exists showing the enhancement of absortion properties by periodicizing the porous layers with embedded scatterers. Periodicity can be imposed in acoustic systems using the so-called Bloch-Floquet theory. There, the layer of infinite extent is limited to a single unit-cell, able to alone describe the full behaviour of the infinitely repeating periodic cell.

- reciprocal wavenumber space, limit the study to the first Brillouin zone covering, $k_1 \in [0, \pi/d]$.
- opening of a Bragg bandgap: conditions of existence in classical metamaterial, can it happen with losses.
- With these complex geometries comes the need for more advanced tools for the discretization of these structures. Introduce finite elements.
- soft solids: aberkane phd

Media periodization

For more complex geometries, we must turn to alternative methods. As the 1D discretization used by the SCM and SAFE and the plane wave formalism used in TMM and the GM is not capable of providing a description of the geometry of inhomogeneous layers, in our case, the effects of the scatterer in the unit cell on wave propagation.

Paragraphe on periodic systems

To introduce periodicity in the configurations we have been investigating up to this section, we will first introduce a few fundamental concepts. Periodic structures and lattices are the central topic of crystallography which has been used in various disciplines such as condensed matter. In wave physics, periodic structures have been shown to introduce interesting effects towards applications for wave control and constitute the cornerstone of metamaterials as was shown in the literature presentation in Section cite sec.

For wave phenomena, periodicity was formalised by the Bloch-Floquet theorem. Taking an infinitely repeating lattice or a crystal, the description of the wave propagation in only one of the so-called unit cell, one spatial period of the total structure, is enough to describe the whole structure. A wave function $\psi(\boldsymbol{x})$ describing the propagation should thus depend on a spatially periodic field, with the same periodicity as the crystal modulated by the Bloch wavenumber k_q, such that,

The scattering of waves by a single unit cell is expressed using a wavefunction that is projected onto an infinite modal basis, the Bloch mode basis.

$$\psi_q(\boldsymbol{x}) = e^{\mathrm{i}\boldsymbol{kx}}. \tag{1.21}$$

Their value at the left boundary is thus related to that at the right boundary modulated by the spatial periodicity and some translation vector. This formalism forms the fundamental framework for periodic structures in acoustics. These structures have been shown to support bands of frequency where waves are unable to propagate: band gaps. These occur at the Bragg frequency described as $d\sin(\theta) = n\lambda$ corresponding to destructive interference for an incident wave impinging on the structure. There are various phenomena and ways to obtain different types of band gaps[**stuff here**].

Overview of the methods

Multiple Scattering Theory is an analytical formalism developed for this purpose and is able to account for the scattering of each of the neighbouring obstacles. Tedious theoretical framework, shown to compute the acoustic response of metaporous[15] and metaporoelastic surfaces[9].

FEM flexible and powerful able to

1.3.2 Dispersion relations and band structure in periodic lattices

Most popular methods for the computation of band structure in 2D periodic systems made of classical lossless materials are the Plane Wave Expansion (PWE) which is semi-analytical. It relies on a formulation... Extended Plane Wave Expansion (EPWE) was generalized to obtain also the evanescent Bloch modes in Ref.[**schalcher2023**], allowing for complex dispersion diagram investigation. The complexity of the system prohibits the use of such semi-analytical methods. Multiple scattering could also be considered. However, it has been successfully used to describe the acoustic scattering of periodically arranged rigid inclusions in equivalent fluid unit cells[15] as well as solid inclusions in a poroelastic unit cell[9]. However, the so-called Schlömilch series that appear in the calculation, from the effect of the scattering of the neighboring cylindrical inclusions appear to be not so manipulable when considering complex longitudinal wavenumbers. Its expression is

$$\sigma_q^X = \sum_{l=1}^{\infty} H_q^{(1)}(k^X ld)\left((-1)^q e^{ik_1^X ld} + e^{-ik_1^X ld}\right), \tag{1.22}$$

and indeed, separating the k_1^X real and imaginary inevitably leads to at least one growing exponential term in the series, leading to it being unable to converge, up to our study.

The wfe as well as FEM has also been widely applied to these geoemetries. Their modularity and versatility lead them to be good candidates for the modelling of our structure.

We face challenges when trying to model this structure using the tools classically used in the literature, that is COMSOL Multiphysics. Although there are built-in tools in the software for "*Eigenfrequency*" studies, the library aims at solving complex frequencies of the system, given a set of parameters for the wavenumber. As it has been discussed earlier in the chapter, the goal of this work is to solve for complex wavenumbers and real frequencies. Even trying a manual implementation of the model in the "*Weak Form PDE*" interface, the tendency of this software to function as a black box introduces a new degree of complexity to this strategy. While FEM remains the soundest strategy for the modeling of the complex inhomogeneous geometry of the unit cell of metaporous surface, we propose to adapt in-house existing frameworks in order to develop numerical tools in this thesis.

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